DEEP LEARNING - FINAL PROJECT

Real vs. AI-Generated Audio Detection

Cost-Sensitive Deep Learning Detection and External Generalization

GROUP 10 | COURSE PRESENTATION | AUGUST 2026

9,879	177,855	0.972	0.682
SONICS songs	10-second clips	Best SONICS F1	Best external F1

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AI Music: Current Policy & Regulatory Landscape

INDUSTRY DIRECTION

AI music is generally allowed — but disclosure, provenance, and abuse prevention are becoming mandatory.

Streaming Platforms

Spotify, Apple Music, YouTube and others generally do not ban AI-generated music. The focus is increasingly on AI metadata/disclosure, fraud prevention, impersonation, and synthetic spam.

Distribution & Detection

Compliance is pushed upstream to distributors to collect metadata. Platforms like Deezer use automated AI detection alongside artist declarations to identify undisclosed synthetic content.

Regulation & Copyright

EU transparency rules and US voice-cloning laws increase legal stakes. Purely AI-generated content receives no US copyright, though human creative contributions remain protected.

BOTTOM LINE: Extremely timely business question The market is moving from voluntary AI disclosure → systematic identification, metadata tagging and enforcement.

Economic Risk & Cost-Sensitive Detection

KEY MODELING ASSUMPTION: MISSING AI CONTENT IS MATERIALLY MORE COSTLY THAN INCORRECTLY FLAGGING HUMAN CONTENT.

False Negative (AI → Human)

- Regulatory & disclosure non-compliance
- Copyright, likeness & voice-cloning exposure
- Synthetic spam & streaming-fraud leakage
- Royalty-pool dilution
- Reputational & partner risk

→ High potential financial & legal cost. Relative cost: 5

False Positive (Human → AI)

- Creator dissatisfaction & reputational friction
- Manual review & operational overhead
- Temporary incorrect labeling
- Need for additional evidence/verification

→ Usually lower & more reversible cost. Relative cost: 1

Cost-Sensitive Evaluation

Cost ratio $C_{FP}=1$, $C_{FN}=5$:

$$\text{Cost} = (FP + 5 \cdot FN) / N$$

Optimal threshold:

$$p^* = 1/(1+5) \approx 16.7\%$$

Standard measures also tracked: Accuracy, F1-Score, Precision, Recall, and confusion-matrix counts — alongside the custom business-cost metric.

Methodology & Design First

1. Rigorous Design Phase

Prior to coding, conducted extensive literature review on synthetic audio detection. Formulated a modular architecture plan through iterative feedback loops with the course lecturer.

2. Modular Interface Lock

Locked strictly defined input/output interfaces. Enabled parallel execution between team members without code collision or merge conflicts.

3. Walking Skeleton

Engineered an end-to-end operational pipeline on a 100-sample subset (50 real, 50 fake) to validate audio batch shapes, loss calculation, and metadata flows before large-scale execution.

Platform & Compute Constraints

Shared Drive & Colab Integration

Designed specifically for execution on free Google Colab tiers without paid GPU nodes. Reusable code was abstracted into separate coding blocks (modules-like), preserving clean notebook environments.

Disconnect Resilience

Implemented aggressive checkpointing for model weights, configs, and training logs to Google Drive. Allowed seamless resumption of model training across free Colab timeout sessions, with archives validated before reuse.

Data Acquisition & Item-Level Split

SONICS Dataset & Acquisition

Benchmark dataset pairing real music with end-to-end synthetic tracks (Suno/Udio). Real audio downloaded via YouTube API, overcoming rate limits and missing track IDs with automated retry scripts.

Split assigned at the song level (seed 42, stratified) BEFORE segmentation — preventing correlated clips from the same song leaking across partitions.

Split	Songs	Real	Fake	10-s clips
Train	6,915	3,500	3,415	124,375
Validation	1,482	750	732	26,616
Test	1,482	750	732	26,864
Total	**9,879**	**5,000**	**4,879**	**177,855**

Artifact Isolation Decision: trained exclusively on SONICS to prevent models from learning cross-source platform artifacts (e.g., YouTube compression vs. Spotify encoding).

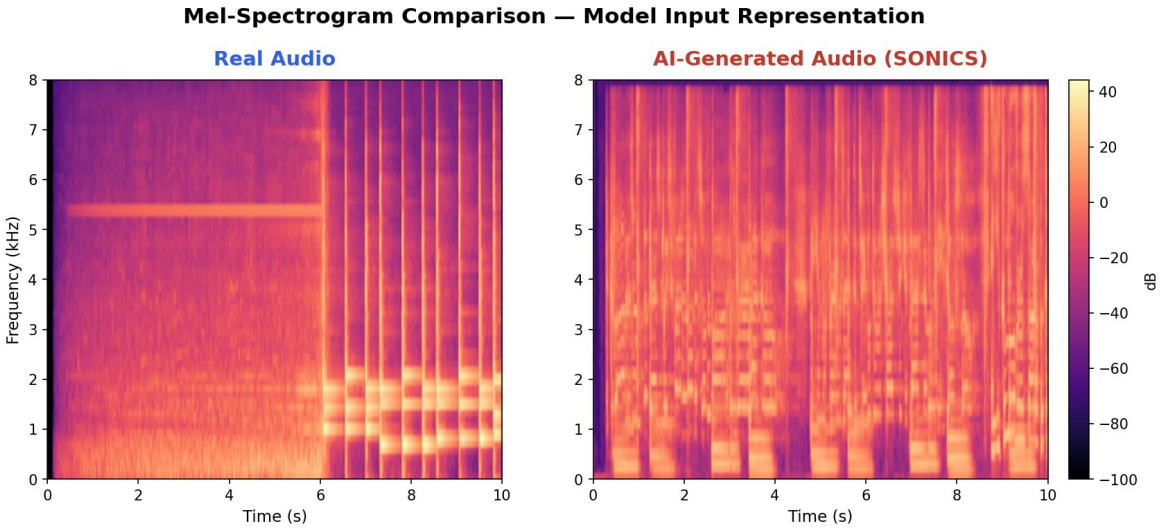
Preprocessing & Song-Level Sampling

Audio Standardization

Standardized raw audio to 16 kHz, mono, 10-second WAV clips. Downsampling acted as implicit noise augmentation. Dynamic on-the-fly augmentation (noise + bitrate) applied strictly during training.

Correcting the Unit of Analysis

The first implementation treated 177,855 clips as a flat pool — overweighting songs with more clips. Fixed: one song = one training example, with 3 candidate clips/song for the 2,100-song HPO subset (1,050 real + 1,050 fake); one fixed clip per song for val/test.



Actual mel-spectrograms from the project's own model input pipeline — real (left) vs. AI-generated (right)

External Evaluation Set: GTZAN + FakeMusicCaps

Executed differently than originally planned: GTZAN was combined with FakeMusicCaps into one balanced external benchmark, so both classes are present and full precision/recall/F1 can be computed — while still analyzing each source separately to expose error type.

External source	Label	Usable recordings	10-s items	Role
GTZAN	Real	735	2,205	OOD real-music / FP stress test
FakeMusicCaps	Fake	2,058 clips	2,058	Unseen generator / FN stress test
Combined	**Both**	—	**4,263**	**Secondary binary generalization benchmark**

Note: some GTZAN source files were corrupted and excluded (735 usable recordings, each split into three 10-second segments).

Three-Tiered Detection Architecture

1. Baseline LSTM

Scratch Architecture. Processes 2D Mel-spectrogram temporal sequences directly via recurrent networks to establish a compute baseline.

2. Mel-CNN

Transfer Learning (PANNs). Leverages frozen Pretrained Audio Neural Networks backbone paired with a trainable linear classification head.

3. Attention Model

Transfer Learning (MERT). Utilizes self-supervised Music Audio Representation Transformer embeddings with a trainable classification head.

Our Hypothesis: Attention beats CNN (and both beat LSTM)

We hypothesize that advanced models generally will beat the LSTM. Since the domain we are working on (music, artistic aspect) includes meaning to timing/pace, we expect that attention beats an image architecture (CNN based).

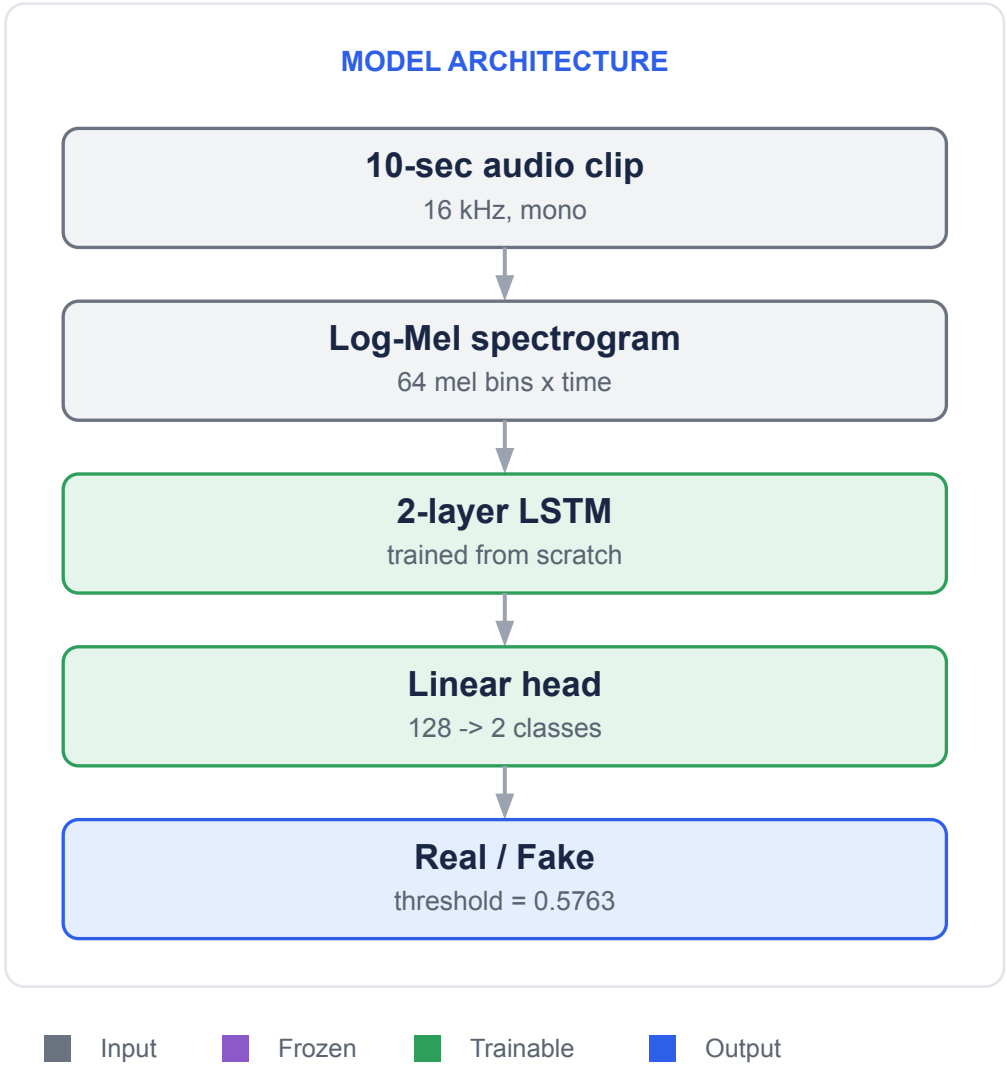
Model 1: Baseline LSTM Architecture

Architectural Principles

Built completely from scratch, no pretrained representations. Extracts 64-bin log-Mel spectrograms as a sequence of acoustic frames.

Final Configuration

2-layer unidirectional LSTM, hidden=128, dropout=0.5, LR=5e-4, 5 epochs. Selected via a manual 3-config grid on validation accuracy (0.924).



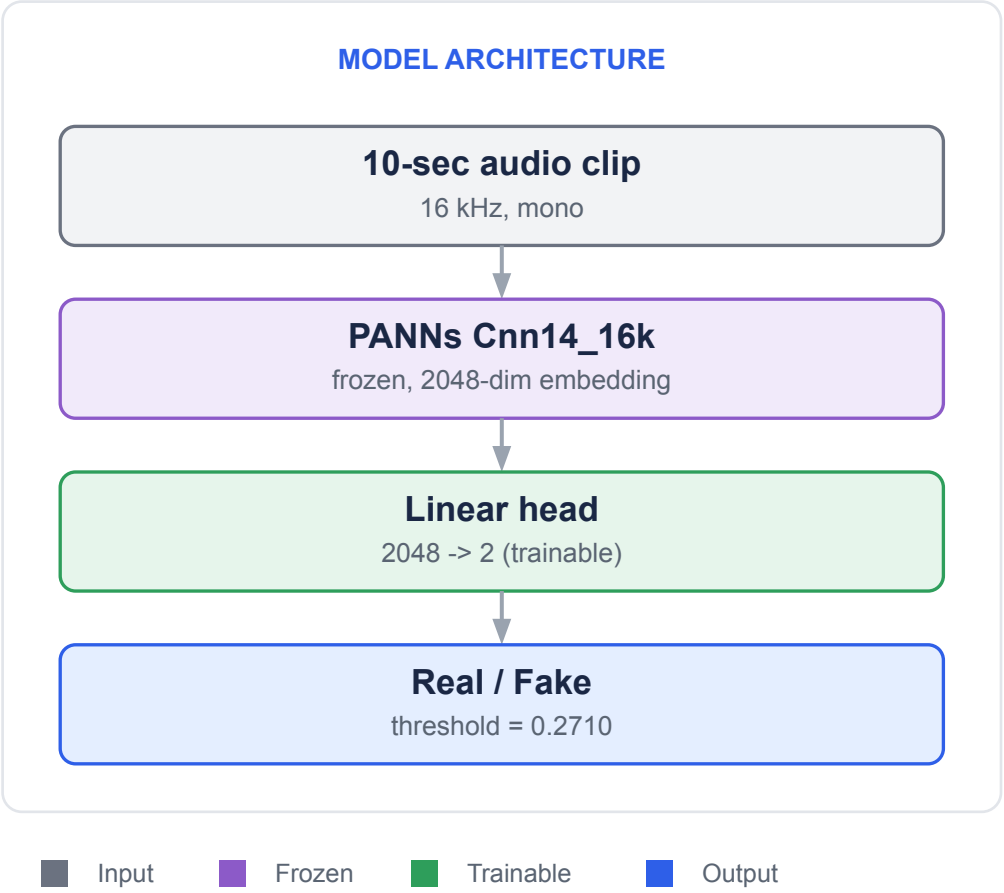
Model 2: Mel-CNN Feature Extractor

Pretrained Backbone

Frozen PANNs Cnn14_16k — explicitly switched from the default 32 kHz checkpoint to match the project's 16 kHz pipeline.

Final Configuration

Dropout=0.3, LR=0.01, 10 epochs. Best validation accuracy 0.910 across a 5-point learning-rate sweep (1e-4 to 0.05).



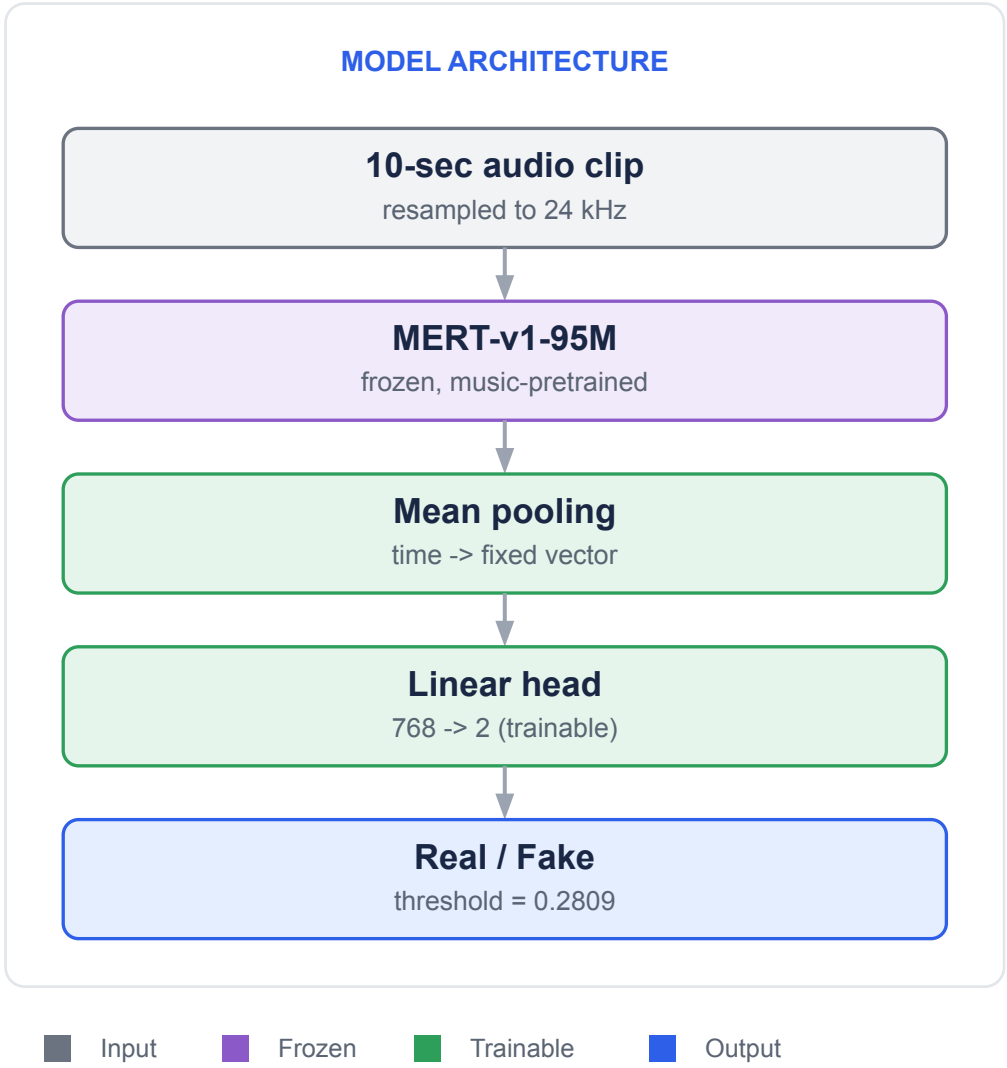
Model 3: Attention Sequence Model

Foundation Model Backbone

Frozen m-a-p/MERT-v1-95M, self-supervised music representation. Waveform resampled 16 kHz → 24 kHz before inference.

Final Configuration

Dropout=0.3, LR=0.01, 10 epochs. Best validation accuracy 0.980 — the strongest in-domain result of the three models.

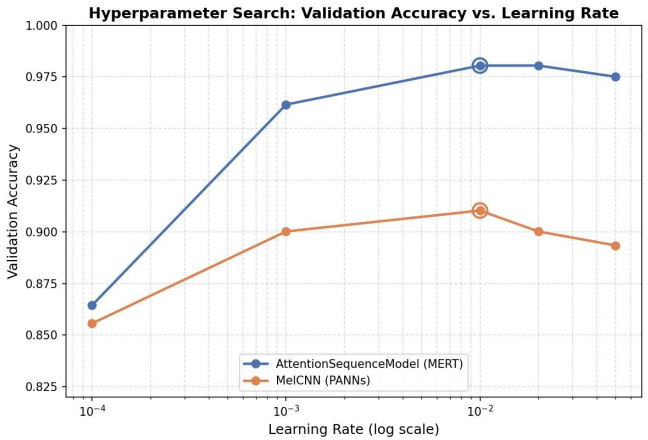


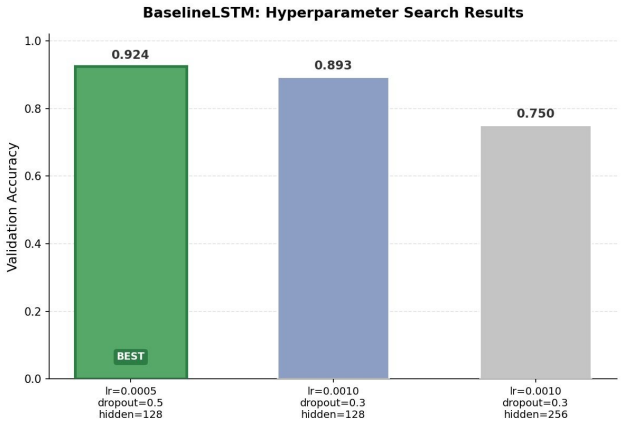
Hyperparameter Search & Threshold Selection

Model	Search tried	Selected config	Best val. acc.
Baseline LSTM	3 configs	Hidden 128, dropout 0.5, LR 5e-4	0.924
Mel-CNN	LR: 1e-4–0.05 (5 pts)	LR 0.01, dropout 0.3	0.910
Attention/MERT	LR: 1e-4–0.05 (5 pts)	LR 0.01, dropout 0.3	**0.980**

Supplementary Optuna Search

A 30-trial search found lower validation cost for Mel-CNN (0.175) and Attention (0.025) than the manual grid — but did NOT translate into better external generalization. Manually tuned checkpoints remain final.





Experimental Results

PRIMARY BENCHMARK — SONICS Test Set

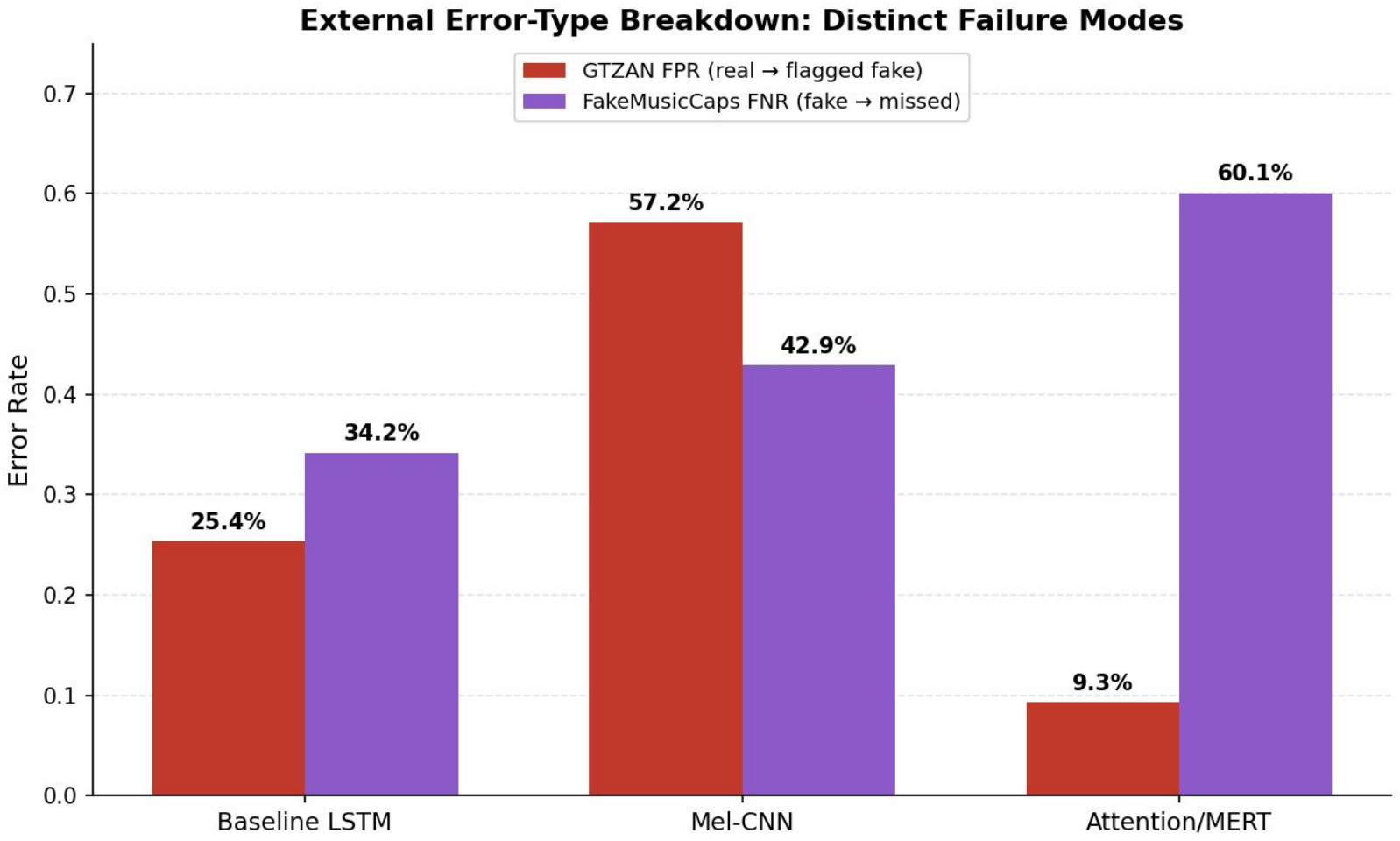
Model	Precision	Recall	F1	Business Cost
Baseline LSTM	0.915	0.990	0.951	0.069
Mel-CNN	0.830	0.967	0.893	0.179
Attention/MERT	**0.948**	**0.997**	**0.972**	**0.034**

SECONDARY — External: GTZAN + FakeMusicCaps

Model	Precision	Recall	F1	Business Cost
Baseline LSTM	**0.708**	**0.658**	**0.682**	**0.956**
Mel-CNN	0.482	0.571	0.523	1.332
Attention/MERT	0.799	0.399	0.532	1.499

Ranking flips under generalization: the in-domain leader (Attention, F1 0.972) drops to a near-tie for last externally (0.532); Baseline LSTM becomes the strongest generalizer (0.682) despite the weakest SONICS score.

Error-Type Breakdown: Distinct Failure Modes



Baseline LSTM: most balanced — best combined external F1 despite not winning either single error axis.

Mel-CNN: high error on BOTH sides — 57.2% real-music FPR, weak overall generalization.

Attention/MERT: conservative — protects real music (9.3% FPR) but misses 60.1% of unfamiliar fakes.

Key Findings & Conclusions

1. Best In-Domain ≠ Best Generalizing

Attention/MERT dominates SONICS (F1 0.972) but collapses externally (F1 0.532, 60.1% miss rate on unfamiliar generators). Baseline LSTM, weakest in-domain, generalizes best (F1 0.682).

2. Distinct, Architecture-Specific Failure Modes

Mel-CNN fails on both real and fake external audio. Attention is conservative (low FPR, high FNR). LSTM is the most balanced, which is why it wins the combined external benchmark.

3. Finer Tuning Increased Overfitting

Optuna found better SONICS validation cost than the manual grid, but exploratory external checks did not show improved generalization — manually tuned checkpoints remain final.

4. Empirical Thresholds Beat the Theoretical One

The cost-derived threshold ($p^*=1/6$) hurt 2 of 3 models — LSTM's precision collapsed to 0.505. Validation-swept, per-model thresholds minimized cost for every model instead.

Final recommendation: Attention/MERT for SONICS-like data; Baseline LSTM for unfamiliar-source robustness — though its external F1 of 0.682 is not yet deployment-ready without further validation.

Limitations & Methodological Caveats

Unequal augmentation across architectures

Dynamic noise/bitrate augmentation reached the LSTM waveform path but not the cached Mel-CNN/MERT embeddings — a comparability gap.

Compute-constrained training subset

HPO and final training used a balanced 2,100-song subset of the 6,915 available training songs, not the full split.

One fixed crop per val/test song

Ensures fair one-song/one-vote comparison, but performance could vary across different excerpts of the same recording.

Optuna comparison is exploratory

Its external check used threshold 0.5, not validation-selected thresholds — excluded from primary model selection.